Name: Bilal Ahmad
Roll No: 054
Class: 5C
Assignment: 2

1. Introduction

In this assignment, I used different computer vision techniques to work with images. First, I detected edges in a Burj Khalifa image. Then I used ORB to find important points and match features between two Burj Khalifa pictures. Finally, I used a Cats vs Dogs dataset to build a simple image classification model.

2. Tools Used

I completed the work in a Kaggle Notebook using Python. I used OpenCV for image processing and ORB, NumPy for numerical data, Matplotlib for displaying results, and scikit-learn for BoVW, k-NN and evaluation metrics.

3. Canny Edge Detection

I loaded the first Burj Khalifa image and converted it into grayscale. I applied Gaussian Blur to reduce noise and then used Canny Edge Detection with thresholds 100 and 200. The output showed the main edges and boundaries of the building.

4. ORB Keypoint Detection

ORB stands for Oriented FAST and Rotated BRIEF. It finds important points, such as corners and detailed areas, in an image. I used ORB with nfeatures=500 to detect keypoints and calculate descriptors. I displayed the keypoints on the Burj Khalifa image.

5. ORB Feature Matching

I used two Burj Khalifa images for feature matching. ORB detected keypoints and descriptors in both images. Then I used a Brute-Force Matcher with Hamming distance to find similar features. The best 50 matches were displayed with lines connecting the matching points.

6. Cats vs Dogs Classification

For classification, I used the Cats and Dogs Image Classification dataset from Kaggle. It contained 557 training images (279 cats and 278 dogs) and 140 testing images (70 cats and 70 dogs). I resized the images to 128 × 128 pixels before extracting features.

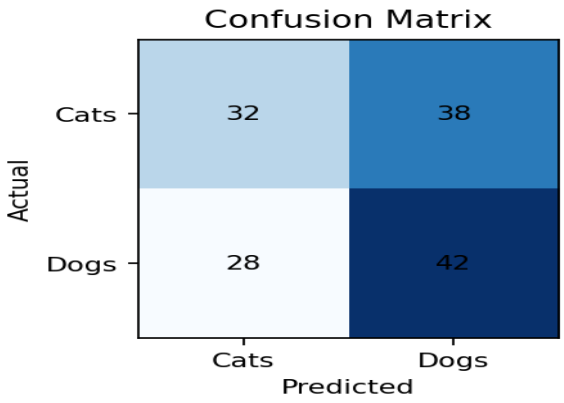
First, I used ORB to extract descriptors from the images. Since each image can have a different number of descriptors, I used Bag of Visual Words (BoVW) to convert them into fixed-size feature vectors. MiniBatchKMeans grouped the descriptors into 50 clusters. Each image was then represented by a histogram of 50 values.

After that, I trained a k-Nearest Neighbors (k-NN) classifier with 5 neighbors. The model used the training images to learn the difference between cats and dogs and then predicted the labels of the test images.

7. Results

The model achieved 52.86% accuracy on the 140 test images. It correctly classified 74 images and incorrectly classified 66 images.

Metric	Cats	Dogs
Precision	0.53	0.53
Recall	0.46	0.60
F1-score	0.49	0.56
Support	70	70



Confusion matrix: [[32, 38], [28, 42]]. The model correctly identified 32 cats and 42 dogs. It confused 38 cats as dogs and 28 dogs as cats.